

TRIASHA SARKAR

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Machine learning engineer and Georgia Tech aerospace MS graduate who builds reliable ML systems from data and evaluation through deployment. Experience spans retrieval, recommendation, GenAI evaluation, time-series modeling, scientific ML, and safety-critical engineering.

EDUCATION

MS, Aerospace Engineering Georgia Institute of Technology, Atlanta, GA <i>Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL)</i>	Aug 2026
MSc, Aerospace Vehicle Design Cranfield University, United Kingdom <i>First Class Honours · Thesis on geometry optimization under FAA constraints</i>	Feb 2023
B.Tech, Aerospace Engineering SRM Institute of Science and Technology, India <i>First Class Honours</i>	Jun 2021

TECHNICAL SKILLS

ML & GenAI: PyTorch, scikit-learn, transformers, RAG, hybrid retrieval, reranking, recommender systems, uncertainty estimation, time-series ML
Systems: Python, SQL, FastAPI, Docker, GitHub Actions, DuckDB, PostgreSQL/pgvector, ONNX, Core ML, Qualcomm QNN
Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Georgia Tech ASDL - I worked across simulation reliability, airline safety retrieval, sustainable aviation, and mobility studies, taking each problem from data review and model design through tested analysis. - I worked with faculty and sponsors to check assumptions, document limits, and deliver models that others could reproduce and use.	May 2025 – Aug 2026
Machine Learning Engineer ALTEN India, embedded at Rolls-Royce - Lifecycle engineers needed clearer signals from multivariate engine data, so I built Python workflows for diagnostics, anomaly detection, predictive maintenance, and mixed-frequency time series. I reviewed the findings with domain experts before they were used. - I turned certification requirements into reproducible analyses and model checks, giving engineering teams a traceable basis for safety-critical decisions.	Oct 2023 – Apr 2025
Data Science Intern Rolls-Royce DataLabs I cleaned aircraft-engine sensor data, designed useful features, and compared predictive and anomaly-detection models. The analysis gave engineers a clearer view of recurring failure patterns.	May – Jul 2021

TECHNICAL PROJECTS

AeroRAG-X Retrieval and LLM Evaluation for Technical Knowledge <i>github.com/triasha72/AeroRAG-X</i> - NASA technical reports are difficult to search and verify, so I built hybrid retrieval, reranking, pgvector search, and citation controls over 3,233 source-preserving chunks. - I added retrieval and answer evaluation, evidence checks, bounded agents, FastAPI services, Docker, tracing, and safe refusals so the system could fail clearly instead of inventing an answer.	2025 – Present
NewsLens Recommendation, Ranking, and Production ML <i>github.com/triasha72/NewsLens</i> - News recommendation can look stronger when future behavior leaks into training. I used chronological splits, personalized TF-IDF ranking, and a training-only fallback, reaching NDCG@10 of 0.366. - I backed the result with paired bootstrap tests and subgroup analysis, then packaged the model with DuckDB, FastAPI, Docker, CI/CD, telemetry, and automated tests.	Jul 2026
EdgeGenBench Scientific ML and On-Device Inference <i>github.com/triasha72/EdgeGenBench</i> - I built an aircraft-design surrogate benchmark with uncertainty estimates, constrained optimization, physics checks, ONNX export, quantization, and mobile inference. - I compared the deployed outputs across Core ML and Qualcomm QNN. The QNN model retained 0.997 mean held-out R^2 on Snapdragon hardware.	Aug 2026 – Present
AeroSynth-Eval Multimodal AI Evaluation <i>github.com/triasha72/AeroSynth-Eval</i> - Synthetic inspection benchmarks were easy to change during experimentation, so I built a fixed 48-scenario VLM evaluation suite with deterministic assets, provenance checks, annotation workflows, and typed evaluator contracts. - I added an MLX-VLM runner with strict validation so experiments cannot silently change the assets, prompts, scenarios, responses, or protected test data.	Aug 2026 – Present

Equity Backtest | Walk-Forward Signal Evaluation

Jul 2026

github.com/triasha72/Equity-Backtest

- I built an expanding-window backtest with transaction costs and kept a record of every specification tested. The apparent momentum edge disappeared after costs and did not meet the significance threshold set before the analysis.

Atlanta Mobility Resilience Digital Twin

2026 – Present

github.com/triasha72/atlanta-mobility-resilience-digital-twin

- I built an OSMnx and NetworkX simulator to test how road closures change travel time and access across Atlanta, including whether some communities carry more of the disruption.

Surrogate Model Learning

2025 – Present

github.com/triasha72/Surrogate-model-learning

- I compared Gaussian process, response-surface, and radial-basis models on nonlinear benchmarks to see where each method breaks down and how sampling changes the error and uncertainty.

Silent Failure Detection in Rocket-Motor Simulations

May – Jul 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- A design sweep was failing without errors or output. I audited the data and source code, built a leakage-safe classifier with 96.8% accuracy across 15,120 simulations, and traced the failures to an uncapped convergence loop.
- I tested the result on unseen geometries and a separate sampling study, then wrote a patch that turns silent hangs into logged convergence errors.

HERO | Source-Aware Retrieval for Airline Safety

2025 – 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- For Delta's HERO program, I worked on retrieval and evaluation methods that keep generated safety answers tied to the technical sources analysts need to review.

GREEN TEA and Project EAGLE | Sustainable Aviation Modeling

Sep 2025 – Apr 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- I built surrogate, uncertainty, demand, and life-cycle models for alternative-fuel and transport studies. The GREEN TEA model is still used by the sponsor.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech

2025 – 2026